RO047

# Speech-Based Stress Detection with Handcrafted and Self-Supervised Features

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## Introduction and Background

Stress is both a physiological and psychological response to a demanding situation, caused by the release of the hormone cortisol into the bloodstream (Obici, 2025). It can be caused by a wide range of factors, such as a hectic work environment, intense cognitive load, interpersonal communication anxiety and physically taxing tasks. There are two types of stress: acute and chronic. Acute stress arises from immediate demands, and can impair judgement, slow response times and reduce overall productivity, while chronic stress results from prolonged exposure, and negatively impacts mental wellbeing and sleep (Kalmbach et al., 2018).

As society continues to become more fast paced and demanding, the incidence of stress related disorders has risen globally, making it an urgent issue to address (Ray, 2022). Consequently, interest in efficient and non-invasive methods of stress detection increased, especially when traditional physiological monitoring methods, such as heart rate sensors and cortisol sampling, are impractical or invasive (Akbar et al., 2019).

Speech has emerged as a promising modality for detecting stress because it is both convenient to capture and carries rich prosodic markers that change in response to physiological and psychological states. Stress-linked increases in voice pitch (F0) have been shown to coincide with cortisol stress responses (Schewski et al., 2025). Further, one's voice may become faster in rate or exhibit more irregularity when stressed (Kappen et al., 2022). To analyse speech for these markers, an effective method of representing audio signals in a way that preserves essential details while remaining computationally manageable is required, because raw audio is too large and noisy for direct use in most machine learning models. (Abdusalomov et al., 2022)

Thus, a variety of **handcrafted features** have been developed over the years. These features, based on expert knowledge of acoustics and speech production, include pitch, energy, and speech rate, among others (Eyben et al., 2016). These handcrafted features have proven effective for detecting stress but often require manual feature engineering and do not adapt to individual differences in speech patterns (Lu et al., 2012). Two widely used feature extraction approaches in speech-related tasks are Mel-Frequency Cepstral Coefficients (MFCCs) and openSMILE. MFCCs provide a compact representation of the short-term spectrum on the mel scale (a scale that mimics how humans perceive sound) and have been widely used for emotion-related speech modelling (Sahidullah et al., 2015). Building on this, openSMILE offers automatic extraction of a broad set of acoustic descriptors, and has shown strong performance in speech emotion classification (Lang & Cao, 2018).

**Self-supervised** learning techniques have been developed in recent years, where models learn to extract features directly from large amounts of unlabelled speech data. These models, such as Wav2Vec 2.0 and other transformer-based architectures, learn representations of speech in an unsupervised manner by predicting parts of the audio signal from its context (Baevski et al., 2020). Unlike handcrafted features, self-supervised models are capable of capturing complex, nuanced relationships in speech that may not be easily identified by traditional methods (Mohamed et al., 2022).

This project aims to compare and evaluate end-to-end pipelines using both handcrafted and self-supervised features for acute stress identification from speech. By comparing these two approaches, we seek to identify the pros and cons of each and explore an ensemble system that combines both handcrafted and self-supervised features. The ultimate goal is to develop a real-time stress detection system that can be used in a variety of applications, such as workplace monitoring (e.g. in air traffic control) and mental health assessments.

## Datasets

Experimentation was carried out on 2 main datasets: StressID and RAVDESS.

StressID is a multimodal dataset with audio, video and physiological signals recorded. It includes emotional video clips, cognitive tasks (math and comprehension), and public speaking to elicit varied stress responses. The dataset comprises 65 participants performing 11 tasks, each attached with a self-rating for relaxation, stress, arousal, and valence (Chaptoukaev et al., 2023). Out of these 11 tasks, 7 were kept for training as only these 7 had a speech segment.

| Task | Baseline | Counting1 | Counting2 | Counting3 | Math | Reading | Relax | Speaking | Stroop | Video1 | Video2 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Included? | ✘ | ✔ | ✔ | ✔ | ✔ | ✔ | ✘ | ✔ | ✔ | ✘ | ✘ |

Table 1: Tasks in StressID that were kept for training and validation.

Secondly, the RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) dataset is a widely used benchmark for emotion-related speech analysis. RAVDESS consists of 24 professional actors, vocalising 2 different short sentences with various emotional expressions (calm, happy, sad, angry, fearful, surprise, disgust, and neutral). (Livingstone & Russo, 2018)

### Audio Standardisation

All recordings from StressID and RAVDESS were resampled to 16 kHz wav file format and converted to single-channel audio to ensure consistent feature extraction across datasets.

### Dataset Labelling

We consider three labeling schemes for StressID:

1) Binary stress classification: self-reported stress on a 0-10 scale was thresholded at 5 (≥5 = stressed, <5 = not stressed).
2) Three-class affect-based labeling: we derived labels using self-reported valence, arousal, and relaxation/stress ratings. Samples were labeled relaxed if valence>5, arousal<5, and relaxation>5; labeled stressed if valence<5, arousal>5, and perceived stress>5; and neutral otherwise.
3) Stress regression: the raw 0–10 self-assessment score was used as a continuous target.

All proposed pipelines were evaluated on the binary labelling scheme, which serves as our primary benchmark. Three-class and regression experiments were conducted using the classical ML baselines only (openSMILE feature vectors with LR/RF/GB/SVM) to provide a controlled comparison.

RAVDESS was used as an auxiliary dataset for emotion-recognition pretraining to learn speech representations that capture prosody and affective cues in our Wav2Vec 2.0 feature extractor. Thus, RAVDESS emotion labels (8 class) were not converted into stress labels.

**Train-Test Split**

We used an 80/20 speaker-independent (the same speaker does not appear in both sets) split across all experiments to enable fair comparison between pipelines. Since StressID is imbalanced (approximately 2/3 stressed vs 1/3 not stressed), we applied random upsampling of the minority class within the training set only. No alteration of the validation dataset was done, thus we adopt balanced accuracy (Bal Acc.) on the validation dataset as the main benchmark of performance across all pipelines.

| Dataset Split | Not Stressed (0) | Stressed (1) |
|---|---|---|
| Training (before upsampling) | 81 | 220 |
| Training (after upsampling) | 220 | 220 |
| Validation | 29 | 41 |

Table 2: StressID label distribution

## Proposed Methods

We compare multiple modeling pipelines that differ in (a) the audio representation (handcrafted acoustic features, self-supervised embeddings, or text) and (b) the downstream classifier (classical ML, sequence models, or Transformer-based text classification).

We evaluate a range of feature representations and classifiers in a controlled setup, and identify the most effective pipeline for stress identification. The code used for all pipelines can be found in Appendix A.

**Pipeline 1: openSMILE (eGeMAPS) → classical ML**

Handcrafted acoustic features were extracted using openSMILE 3.0, using the eGeMAPSv02 feature set (88 dimensions). From this set, we selected the 37 prosodic features (see Appendix B) most relevant to stress-related vocal variation. We trained 5 classical classifiers using scikit-learn on these 37 prosodic features: Logistic Regression, Random Forest, Gradient Boosting, and Support Vector Machines (SVM Default and Tuned using grid search over hyperparameters on training set).

**Pipeline 2: openSMILE (windowed) → LSTM**

To model temporal dynamics, each recording was split into 10-second windows with a 5-second hop.

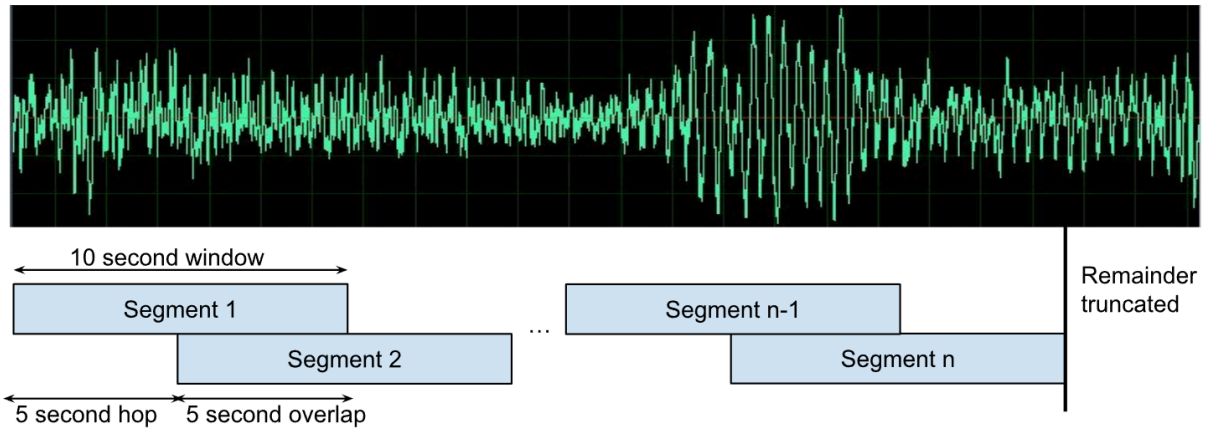


Figure 1: Audio segmentation method

Then, openSMILE feature extraction using the same method as pipeline 1 was performed on each audio segment. Next, the audio segments were sequenced and merged with their dataset labels, forming an n×37 matrix used for model training and validation.

A unidirectional LSTM (3 layers, 0.2 dropout probability, binary cross-entropy loss) was trained to produce a single recording-level stress probability (sigmoid output). With this, temporal information is modeled by the LSTM directly, instead of independently classifying windows and aggregating their predictions.

**Pipeline 3: Wav2Vec 2.0 fine-tuned audio classifier**

Firstly, we utilise RAVDESS to fine-tune the feature extractor of a Wav2Vec 2.0 model (jonatasgrosman/wav2vec2-large-xlsr-53-english). We report the accuracy and f1 score on validation splits of the model for RAVDESS in Table 5 (See Appendix C).

Then, we initialized the Wav2Vec2.0 model previously fine-tuned on RAVDESS, and used it as a feature extractor for downstream stress detection on StressID by attaching a classification head (with dense layer + dropout + output projection) on top of it. During training, we froze feature encoders and optimized the classification head to predict the target labels.

**Pipeline 4: ASR → BERT text classifier**

Together with the previous strategies, which relied on statistical representations of speech, we also explore stress detection on the spoken words themselves. We transcribed each audio file using a pretrained ASR model (facebook/wav2vec2-base-960h) with its corresponding processor, which has a word error rate of 8.6% (AI at Meta, n.d.). Then, we fine-tuned a Transformer encoder (bert-base-uncased) for binary text classification. The model predicts whether a sentence contains stress-related content, and is trained on two labelled datasets: SAD (Mauriello et al., 2021) and Dreaddit (Turcan & McKeown, 2019). We report the accuracy and f1 score of the BERT model on detecting stress content in sentences in SAD and Dreaddit in Table 6 (See Appendix D)

| Sentence text | Prediction | Notes |
|---|---|---|
| "I'm so anxious about the upcoming deadline, I can't sleep." | Stressed (Confidence: 99.98%) | Examples of accurate classification |
| "The weather is beautiful today, and I feel great." | Not Stressed (Confidence: 99.70%) | |
| "I have an exam tomorrow but it's easy. Likely not that hard since it's just an aptitude test." | Stressed (Confidence: 99.98%) | Example of misclassification (false positive) |

Table 3: Predictions of BERT transformer encoder on different input sentences

In qualitative error analysis (Table 3), it can be observed that the classifier performs well on explicit stress statements but often fails on indirect or context-dependent expressions of stress (e.g., sarcasm). It is also important to note that many StressID tasks contain minimal informative lexical content: in counting tasks, participants only recited numbers; in the Stroop task, participants only read colour words aloud). Hence, the model had limited semantic cues to exploit beyond pure transcription noise.

**Ensemble**

Finally, we used a weighted probability ensemble over the 4 pipelines. Ensemble weights were selected to be proportional to balanced accuracy of each pipeline on the training set. Then, the final prediction was computed as a weighted mean of model probabilities.

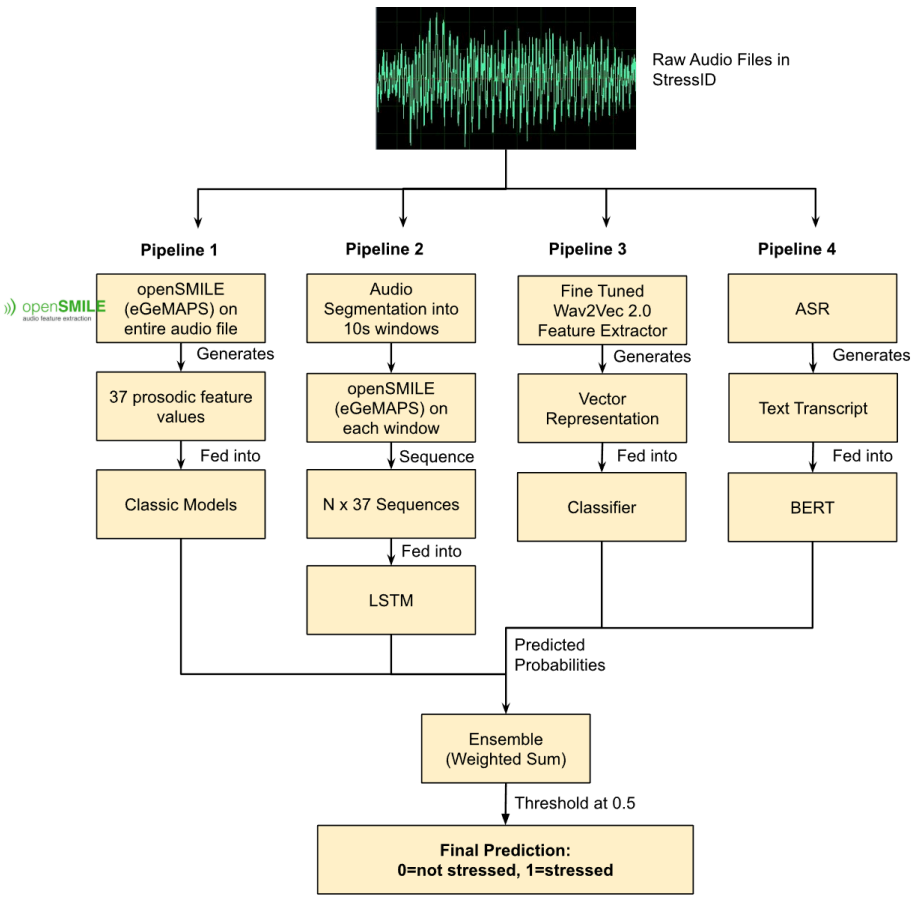


Figure 2: Flowchart of stress prediction

## Results

| Pipelines | Representation | Model | Bal Acc | F1 (macro) | AUC (ROC-AUC) | Precision (stressed) | Recall (stressed) |
|---|---|---|---|---|---|---|---|
| Baseline (majority class) | - | - | 0.50 | 0.37 | 0.50 | 0.59 | 1.00 |
| openSMILE (eGeMAPS) → classical ML | eGeMAPSv02 (37d) | SVM (RBF, tuned) | 0.64 | 0.64 | 0.50 | 0.64 | 0.64 |
| | | GB (Tuned) | 0.60 | 0.58 | 0.62 | 0.69 | 0.60 |
| | | SVC (Baseline) | 0.52 | 0.50 | 0.62 | 0.52 | 0.52 |
| | | **Logistic Reg.** | **0.69** | **0.69** | **0.76** | **0.69** | **0.69** |
| | | Random Forest | 0.59 | 0.55 | 0.55 | 0.75 | 0.59 |
| openSMILE (windowed) → LSTM | Windowed eGeMAPS (37d) sequence | LSTM | 0.66 | 0.66 | 0.73 | 0.69 | 0.83 |
| Wav2Vec2 fine-tuned classifier | Wav2Vec2 embeddings (finetuned) | Wav2Vec2 classifier | 0.64 | 0.63 | 0.67 | 0.67 | 0.93 |
| ASR → BERT text classifier | ASR transcript | BERT-base classifier | 0.44 | 0.35 | 0.55 | 1.00 | 0.05 |
| **Ensemble (weighted)** | **Probabilities from each pipeline** | **Weighted mean** | **0.70** | **0.70** | **0.81** | **0.74** | **0.78** |

Table 4: Two class label results

The performance of all pipelines on StressID is shown in Table 4. All of the pipelines yielded higher than baseline accuracies, except for the ASR→BERT text pipeline (due to the low lexical content of StressID recordings). Amongst the classical models, Logistic Regression obtained the highest balanced accuracy of 69%. Surprisingly, Wav2Vec2.0 and openSMILE(windowed) →LSTM did not do as well as the classic ML models. Their high recall values and low precision values reveal that they predict stress too often, even though the training dataset has been upsampled to balance the classes.

We hypothesise that the relatively small amount of labelled StressID speech available for fine-tuning higher-capacity models, led to unstable decision thresholds and overfitting to stress-like acoustic artefacts. In addition, the windowed LSTM approach can amplify local fluctuations in prosody, causing the model to favour the stressed class when short segments resemble stress even if the overall clip is not stressed.

The ensemble gave the best results, achieving the highest balanced accuracy of 70% and ROC-AUC of 0.81 by combining complementary strengths from multiple pipelines. This is an improvement on the traditional baseline models provided by StressID which obtained 62% (Chaptoukaev et al., 2023). In particular, it balances the robust generalisation of the classical handcrafted-feature models with additional signals captured by the deep representations, leading to improved overall stability across participants and tasks.

Table 7 (see Appendix E) reports the results of three-class labelling strategy. Random forest obtains the highest balanced accuracy of 54%. Although its accuracy is higher than a random baseline, it shows that openSMILE prosodic features do not have the granularity to reliably distinguish three classes (relaxed, neutral, stressed) over binary class.

Table 8 (see Appendix F) reports the results of the regression labelling strategy. Random forest again obtains the best results, with a notable 44% of predictions being within 1 point of true 0–10 score. This shows the viability of using a regression label instead of a class label for modelling stress.

## Conclusion

This work has shown that prosody is a strong predictor for stress, with models achieving the highest performance when trained on handcrafted features, indicating that pitch, energy, and voice quality speech markers can indeed be used for acute stress identification. In addition, self-supervised features are comparable to handcrafted features in terms of capturing stress-related information, but notably did not consistently surpass the strongest classical baselines under the current data constraints, suggesting that performance is limited more by dataset size and label noise than by representation capacity alone. The ensemble of all pipelines yielded significant improvements over traditional baselines provided in StressID.

Quality and quantity of recordings in StressID is an important limitation in our study. With only roughly 6 hours of recorded speech, it is insufficient to train a model that can generalise tiny details across different speakers to predict stress. This limitation is particularly pronounced for high-capacity deep models (Wav2Vec2.0), where fine-tuning can lead to unstable thresholds or overfitting, especially under speaker-independent validation. Furthermore, the labels in StressID are based on self-assessed values of stress which are subjective to each participant, leading to high data variability. thus work has to be done on utilising physiological data to create a fairer and more accurate measurement of stress, rather than relying on self-assessed values.

A possible method moving forward could be taking the percentage difference between the mean of recorded baseline and stressor-induced ECG/EDA/RR as a measurement. This would allow stress to be quantified relative to each participant’s own baseline, reducing inter-speaker variability and improving label consistency. Correlation between self-assessed values and physiological data can also be explored to ensure validity of method.

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# Appendix A

Github repository (https://github.com/StifflenTwee/RO047) for code used for data extraction, processing, and machine learning models.

## Appendix B – List of the 37 prosodic features used for openSMILE extraction

F0semitoneFrom27.5Hz_sma3nz_amean
F0semitoneFrom27.5Hz_sma3nz_stddevNorm
F0semitoneFrom27.5Hz_sma3nz_percentile20.0
F0semitoneFrom27.5Hz_sma3nz_percentile50.0
F0semitoneFrom27.5Hz_sma3nz_percentile80.0
F0semitoneFrom27.5Hz_sma3nz_pctlrange0-2
F0semitoneFrom27.5Hz_sma3nz_meanRisingSlope
F0semitoneFrom27.5Hz_sma3nz_stddevRisingSlope
F0semitoneFrom27.5Hz_sma3nz_meanFallingSlope
F0semitoneFrom27.5Hz_sma3nz_stddevFallingSlope
loudness_sma3_amean
loudness_sma3_stddevNorm
loudness_sma3_percentile20.0
loudness_sma3_percentile50.0
loudness_sma3_percentile80.0
loudness_sma3_pctlrange0-2
loudness_sma3_meanRisingSlope
loudness_sma3_stddevRisingSlope
loudness_sma3_meanFallingSlope
loudness_sma3_stddevFallingSlope
jitterLocal_sma3nz_amean
jitterLocal_sma3nz_stddevNorm
shimmerLocaldB_sma3nz_amean
shimmerLocaldB_sma3nz_stddevNorm
HNRdBACF_sma3nz_amean
HNRdBACF_sma3nz_stddevNorm
alphaRatioV_sma3nz_amean
alphaRatioV_sma3nz_stddevNorm
hammarbergIndexV_sma3nz_amean
hammarbergIndexV_sma3nz_stddevNorm
slopeV0-500_sma3nz_amean
slopeV0-500_sma3nz_stddevNorm
slopeV500-1500_sma3nz_amean
slopeV500-1500_sma3nz_stddevNorm
alphaRatioUV_sma3nz_amean
hammarbergIndexUV_sma3nz_amean
loudnessPeaksPerSec

## Appendix C

| Representation | Model | Class | Precision | Recall | f1-score | Support |
|---|---|---|---|---|---|---|
| Wav2Vec2 embeddings (not finetuned) | Wav2Vec2 Classifier | angry | 0.97 | 0.95 | 0.96 | 38 |
| | | calm | 0.84 | 0.95 | 0.89 | 39 |
| | | disgust | 1.00 | 0.95 | 0.97 | 38 |
| | | fearful | 0.90 | 0.90 | 0.90 | 39 |
| | | happy | 0.95 | 0.95 | 0.95 | 38 |
| | | neutral | 0.77 | 0.89 | 0.83 | 19 |
| | | sad | 0.88 | 0.76 | 0.82 | 38 |
| | | surprised | 0.97 | 0.97 | 0.97 | 39 |
| | | accuracy | - | - | 0.92 | 0.92 |
| | | macro avg | 0.91 | 0.92 | 0.91 | 288 |
| | | weighted avg | 0.92 | 0.92 | 0.92 | 288 |

Table 5: RAVDESS Wav2Vec 2.0 results

## Appendix D

| Task | Class | Precision | Recall | f1-score | Support |
|---|---|---|---|---|---|
| BERT model on datasets SAD + Dreaddit | Not stressed | 0.78 | 0.72 | 0.75 | 345 |
| | Stressed | 0.94 | 0.96 | 0.95 | 1593 |
| | accuracy | - | - | 0.92 | 1938 |
| | macro avg | 0.86 | 0.84 | 0.85 | 1938 |
| | weighted avg | 0.91 | 0.91 | 0.91 | 1938 |

Table 6: Accuracy of BERT model in detecting stress from sentences.

# Appendix E

| Pipeline | Repres-entation | Model | Bal Acc | F1 (macro) | AUC (ROC-AUC) | F1 (relaxed) | F1 (neutral) | F1 (stressed) |
|---|---|---|---|---|---|---|---|---|
| Baseline (majority class) | - | - | 0.3333 | 0.2092 | nan | 0 | 0.6275 | 0 |
| openSMILE (eGeMAPS) → classical ML | eGeMAP Sv02 (37d) | SVM (RBF, tuned) | 0.4362 | 0.4015 | 0.5816 | 0.4348 | 0.2553 | 0.5143 |
| | | GB (Tuned) | 0.496 | 0.4878 | 0.6121 | 0.5882 | 0.3182 | 0.557 |
| | | SVC (Baseline) | 0.4058 | 0.3776 | 0.5928 | 0.3448 | 0.3396 | 0.4483 |
| | | Logistic Reg. | 0.4939 | 0.4588 | 0.626 | 0.3871 | 0.4074 | 0.5818 |
| | | Random Forest | 0.5472 | 0.5368 | 0.5823 | 0.6316 | 0.4 | 0.5789 |

Table 7: Three class results

# Appendix F

| Pipeline | Representation | Model | MAE | RMSE | % within ±1 (percentage of predictions within 1 point of the true 0–10 score) |
|---|---|---|---|---|---|
| Baseline (mean) | - | - | 1.7188 | 1.9843 | 28.57 |
| openSMILE (eGeMAPS) → classical ML | eGeMAPSv02 (37d) | SVM (RBF, tuned) | 1.6706 | 1.9216 | 28.57 |
| | | GB (Tuned) | 1.3877 | 1.7038 | 42.86 |
| | | SVC (Baseline) | 1.5339 | 1.8728 | 38.57 |
| | | Logistic Reg. | 1.6804 | 2.1008 | 32.86 |
| | | Random Forest | 1.3586 | 1.7339 | 44.29 |

Table 8: Regression label results